

# Hamza Kyamanywa

## Senior Fullstack Software Engineer

React Native • Node.js • MongoDB • AWS • AI/LLM

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### Professional Summary

Senior Fullstack Engineer with 4+ years building scalable B2C mobile and web applications at a leading Korean EdTech company (50K+ MAU). Expertise in **React Native, Node.js, MongoDB, and AWS** with proven track record delivering high-impact features that **increased user retention by 40%** and drove **25% premium subscription growth**. **Fluent in Korean (TOPIK 5)** and English, with strong AI/LLM engineering capabilities. Award-winning engineer (**2x Hero Awards**) with rapid promotion from Level 2 to Level 4 in 18 months.

### Technical Skills

**Mobile & Frontend:** 📱 **React Native (iOS/Android)**, ReactJS, Next.js 15, React 19, TailwindCSS, Lexical Editor, EditorJS

**Backend & APIs:** 🌐 **Node.js**, 🚀 **FastAPI**, Django, Express.js, RESTful APIs, WebSocket, Socket.io

**Databases:** 🟢 **MongoDB**, 🐉 PostgreSQL, MySQL, Redis, Firebase, Supabase, Pinecone (Vector DB)

**Cloud & DevOps:** 🍌 **AWS (Lambda, ECS, S3, EC2, EventBridge)**, Docker, GCP, CI/CD, Serverless Architecture

**Cross-Platform:** React Native, Electron, Chrome Extensions (Manifest V3), PWA, Desktop Apps

**AI/ML:** LangChain, LangGraph, OpenAI API, Whisper, RAG, Vector Embeddings, Prompt Engineering

**Languages:** 🇰🇷 **Korean (Fluent - TOPIK 5)**, 🇺🇸 English (TOEFL 111, IELTS 7.5), 🇸🇩 Luganda (Native)

**Programming:** 🐍 Python, 🟡 JavaScript / 🟦 TypeScript, 🟢 C++, SQL

### PROFESSIONAL EXPERIENCE

#### Technical Founder

**HoverNotes (AI Video Note-Taking SaaS) | Remote, United States | Jul 2025 - Sep 2025**

- Founded and led technical development of **privacy-first AI video note-taking platform** and grew it to over from \$0 to \$1100 MRR in 4 months
- Full-Stack Architecture:** Built three-tier system (Chrome Extension + Next.js 15 Web App + Python AI Backend) with React 19, TypeScript, FastAPI, LangGraph agents, and Docker
- Privacy Innovation:** Implemented local-first architecture with direct Obsidian vault integration and zero cloud storage
- Universal Video Compatibility:** Engineered advanced video detection working across YouTube, Udemy, LinkedIn Learning, Bilibili, and all major platforms
- AI Integration:** Built multi-modal processing with real-time streaming note generation using LangGraph reasoning agents
- Global Reach:** Developed 45+ language localization with culturally-adapted AI note generation and phased market rollout strategy
- Tech Stack:** Next.js 15, React 19, TypeScript, PostgreSQL, Drizzle ORM, Stripe, Supabase, Tailwind CSS, Docker, Python, FastAPI, Docker, Langchain, AWS (EC2, Route53, S3, Lambda)

#### Senior Full Stack Engineer (Freelance)

**GoGymi (EdTech SaaS Platform) | Remote, Switzerland | Jan 2025 - Sep 2025**

- Led full-stack development of TexTutor, an AI-powered educational SaaS platform for Swiss teachers and students
- Full-Stack Architecture:** Built complete platform using **Next.js 15, React 19, PostgreSQL, TypeScript** with comprehensive responsive design
- Billing & Monetization:** Implemented **Stripe subscription system** with multi-tier plans, free trials, and automated workflows
- Multi-Language Platform:** Delivered internationalization supporting 5 languages (EN/DE/ES/FR/IT) with localized billing
- Authentication & Security:** Built complete user management with email verification, password reset, and secure session handling
- Rich Features:** Implemented Lexical editor with collaborative features, LiveBlocks integration, and PDF export capabilities
- Tech Stack:** Next.js 15, React 19, TypeScript, PostgreSQL, Drizzle ORM, Stripe, Supabase, Tailwind CSS, Docker

#### AI Engineer & Team Lead (Level 4)

## Bebridge Inc. (Slid - EdTech Platform, 50K+ MAU) | Seoul, South Korea | Jan 2024 - Jun 2025

- Led AI innovation transforming Slid into industry-leading AI-powered educational platform with 50,000+ monthly active users
- **Mobile App Development:** Spearheaded **React Native iOS/Android app launch** with server-side video processing, increasing user base by 21% and mobile retention by 40%
- **AI Auto-Notes Pioneer:** Built end-to-end LLM/Whisper-based auto-notes from video, increasing videos opened by 12% and overall retention by 25%
- **Cost Optimization:** Achieved 60% API cost reduction through intelligent buffering and multi-provider optimization
- **Server-Side Video Processing:** Developed Ffmpeg-based screenshot solution on AWS Lambda overcoming mobile platform limitations
- **AI Agent Systems:** Built LangGraph multi-agent pipeline for transcript correction with custom reasoning tools
- **Backend Innovation:** Reverse-engineered YouTube video delivery for direct URL access, enabling mobile processing at scale
- **Tech Stack:** React Native, Python, FastAPI, AWS Lambda, EventBridge, S3, Ffmpeg, LangGraph, OpenAI/Whisper APIs

## Senior Software Engineer (Level 3) - R&D Team

### Bebridge Inc. | Seoul, South Korea | Jun 2023 - Dec 2023

- **AI Sliddy Chat (Industry First):** Built AI chat for notes before Notion AI, using LangChain, Pinecone vector DB, and streaming responses
- **Semantic Search:** Implemented RAG-based search increasing session duration by 25%
- **LLM Video Summarization:** Built scalable event-driven cloud engine achieving 1,500 MAU and 25% weekly retention
- **Cost Reduction:** Reduced summarization costs by 40% through result caching and API optimization
- **Performance:** Improved summarization speed by 66% via parallel processing and lazy evaluation
- **Universal Export:** Built multi-format export (Word, PDF, Markdown, Notion) with 96% accuracy, reducing support by 75%
- **Open Source:** Contributed to LangChain, Vercel AI SDK, and Quivr
- **Recognition:** Awarded Hero Award 2023, promoted to Level 4

## Full Stack Engineer (Level 2)

### Bebridge Inc. | Seoul, South Korea | Jun 2022 - May 2023

- **Real-Time Transcription:** Pioneered WebRTC → Socket.io → Google Cloud Speech STT, increasing premium subscriptions by 51%
- **Stack Overflow Impact:** Solution helped 5,000+ developers with WebRTC/STT implementation
- **Cost Optimization:** Achieved 90% transcription cost reduction through provider evolution (Whisper → Google → Groq)
- **Authentication System:** Designed cross-platform auth using AWS Cognito for Web, Chrome Extension, Mobile (React Native), and Desktop
- **Payment Integration:** Built **Stripe + Payple** multi-gateway system with 35% checkout improvement and 98%+ success rate
- **Cross-Platform Development:** React Web App, **React Native Mobile**, Electron Desktop, Chrome Extension (200+ commits)
- **Performance:** Reduced load time by 57% through lazy loading and optimization
- **Open Source:** Contributed to EditorJS and Redux Toolkit
- **Recognition:** Awarded Hero Award 2022, promoted to Level 3

## EDUCATION

### Bachelor of Science in Electrical and Electronics Engineering

#### Korea University, Seoul | March 2019 - February 2023 | GPA: 4.02/4.5 (Magna Cum Laude)

- **Publication:** "Responsible Music Genre Classification using Interpretable Model-agnostic Visual Explainers" in Springer Nature Computer Science Journal (4th ICDSA 2023)
- **Scholarships:** Global Korea Scholarship (2018-2023), Ugandan Government Scholarship (2017)

## AWARDS & RECOGNITION

- **Bebridge-Slid Hero Award** (2 consecutive years: 2022, 2023) - Top performer company-wide
- **Rapid Promotion:** Level 2 → Level 3 (2022) → Level 4 (2023) in 18 months
- **Korea University Academic Excellence Award** (July 2020)
- **Korea University Dean's List** (3 semesters: 2019-2020)

## OPEN SOURCE CONTRIBUTIONS

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- **LangChain** (118K ★) - Framework for LLM applications
  - **EditorJS** (25.1K ★) - Block-style text editor
  - **Redux Toolkit** (11.1K ★) - Official Redux toolset
  - **Vercel AI SDK** (18.7K ★) - AI-powered streaming UIs
  - **MemGPT** (18.9K ★) - LLM agents with long-term memory
  - **Pytube** (13.5K ★) - YouTube video downloading library
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## REFERENCES

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### Junghyun Park

**CEO & Founder, BEBRIDGE INC.**

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*Direct supervisor during tenure at Bebridge Inc. (Slid platform)*

### Beelee Chua

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*Academic supervisor and research mentor*